

Future Direction Report

SIT708 Pass Task 7.1 – Lost and Found App

The current app stores adverts locally in SQLite [1], which means a lost item posted on one device is invisible to anyone else. The most impactful next step is migrating to a shared backend so reports become discoverable across users. Firebase Firestore [2] is a practical choice because it offers real-time synchronisation, authentication, and storage for image files without requiring custom server infrastructure. Each advert would carry a user identifier, enabling owners to manage only their own postings while browsing reports from the wider community.

A second direction is location intelligence. The wireframe already captures a textual location, but resolving this through the Geocoding API [3] into latitude and longitude unlocks map-based browsing with the Google Maps SDK and proximity search such as “items lost within two kilometres of my current position.” Combined with push notifications via Firebase Cloud Messaging [4], users could be alerted when a newly posted found item matches a category and area they previously reported as lost.

Finally, trust and safety deserve attention. Verified contact through one-time SMS codes, in-app messaging rather than exposed phone numbers, and a simple reporting mechanism for fraudulent listings would make the platform safer at scale. Together these directions transform a single-user SQLite prototype into a community-driven service capable of meaningfully reuniting people with what they have lost.

References

- [1] "Save data using SQLite," Android Developers, 2026. [Online]. Available: <https://developer.android.com/training/data-storage/sqlite>. [Accessed: 28-May-2026].
- [2] "Cloud Firestore documentation," Firebase, Google, 2026. [Online]. Available: <https://firebase.google.com/docs/firestore>. [Accessed: 28-May-2026].
- [3] "Geocoding API overview," Google Maps Platform, Google, 2026. [Online]. Available: <https://developers.google.com/maps/documentation/geocoding/overview>. [Accessed: 28-May-2026].
- [4] "Firebase Cloud Messaging," Firebase, Google, 2026. [Online]. Available: <https://firebase.google.com/docs/cloud-messaging>. [Accessed: 28-May-2026].